Ridgeline: Lift Line Laboratory

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Abstract and Context

Methodology

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Every skier plays some version of the same game: “How do I maximize my enjoyment of the mountain?” For many, the answer is straightforward: maximize the time actually spent skiing.

In recent years, ski resorts have faced increasing criticism over long lift lines, unreliable operations, and the perceived inadequacy of the experience relative to rising ticket prices. Yet lift-line congestion is not simply a function of the number of skiers on a mountain or the capacity of its infrastructure. It is also influenced by how skiers distribute themselves across the resort.

Consider a skier arriving at the bottom of a run and encountering a long lift line. They face a simple decision: enter the queue immediately or travel to another lift in hopes of finding a shorter wait. The alternative, however, carries uncertainty. While the skier can observe the queue directly in front of them, their knowledge of other lift lines may be incomplete or outdated.

Apply that behavioral pattern to hundreds of skiers independently, each making similar decisions attempting to maximize their own time on the mountain. Their collective behavior may produce congestion patterns that no individual skier intended.

This project investigates how heterogeneous skier decision-making influences lift-line congestion and queue stability. Using a simplified system of population-level differential equations and an agent-based simulation of a ski resort, it examines how individual terrain preferences, remembered queue lengths, and access to alternative lifts affect the distribution of skiers across the mountain.

Note: This paper uses the term “skier” as an analog for both skiers and snowboarders in addition to others who use the mountain such as monoskiers, and snowfeet

Methodology

The model is constructed progressively, beginning with a generalized mathematical representation of an individual skier and concluding with a population-level description of lift-line congestion; first defining the skier's objective, introducing empirically motivated heterogeneity, then establishing class-specific behavioral constraints. After which, movement through the resort and individual decision-making under incomplete information are both formalized.

For the initial descriptions, lift waiting times are assumed constant. This assumption is subsequently relaxed to account for the interaction of individual decisions, allowing queue populations to evolve dynamically. The resulting equations provide the foundation for a stochastic agent-based simulation.

2.1 General Skier Model

Consider an individual skier *i* participating in a finite operating day of duration *H*. Assume that the skier seeks to maximize the utility derived from time spent skiing, subject to time consumed by lift queues, uphill transportation, and transit between lifts. This can be expressed as such:

$$T_{skiing} = H - \left(\sum_{x=0}^{y} T_{queue} + \sum_{x=0}^{y} T_{uphill} + \sum_{j=0}^{l} T_{transit}\right)$$

Moreover, the skier's available time is constrained by the amount of time they spend on the resort, modifying *H*. Henceforth *H*, when unmodified, can be interpreted as the upper bound of time that an individual has to ski the resort. The general skier model makes no assumptions

regarding terrain preferences, ability, or willingness to change routes. These characteristics are introduced through class-specific restrictions.

2.2 Empirical Motivation for Skier Heterogeneity

A homogeneous population is insufficient to represent differences in terrain accessibility, route preference, and willingness to relocate.

Schlemmer and Peters published a research paper depicting several unique skier phenotypes. These observations motivate the introduction of heterogeneous skier populations. The present model distinguishes four behavioral classes: all-mountain skiers, vacationers, lappers, and mixed skiers derived from said article[1].

For the purposes of this paper, the proportion of the resort population belonging to classes is as follows:

All-Mountain Skiers	25%
Vacationers	35%
Lappers	15%
Mixed Skiers	25%

Note that these population percentages are assumed, not empirically estimated. The class proportions are treated as model parameters, with their values stated explicitly and varied where appropriate.

All-Mountain Skiers can ski any terrain with no special behavior excluding the baseline minimization. Vacationers are limited to skiing only blue and green difficulty runs with no special behavior excluding the baseline minimization. Lappers are limited to the type of terrain

admissible to their sub-division (Park, Race, General) choosing one run, or in the case of the Park-Lapper two, runs to repeat throughout the day. Mixed Skiers initially ski the entire mountain (40% of the day), before repeating the two runs which had the lowest empirical wait times for the rest of the day (60%).

2.3 Class-Specific Skier Models

The generalized objective in Section 2.1 is modified through class-dependent terrain preferences and admissible route choices.

All-Mountain Skiers	May access all modeled terrain and change lifts in response to perceived waiting-time advantages.
Vacationers	Are restricted to terrain compatible with their skiing ability. Their access to an additional lift depends on the resort's operating configuration.
Lappers	Prefer a particular run or group of runs and remain committed to their associated lift, irrespective of congestion elsewhere.
Mixed	Initially explore the mountain before developing preferences for particular runs. They retain the ability to relocate when waiting-time advantages justify doing so.

For the purposes of the variable assignment, let $A_i(t)$ denote the set of lifts accessible to skier i.

2.4 Mountain Topology and Temporal Constraints

The resort is represented as a network consisting of twenty ski runs and four lifts, with three or four lifts operational according to the experimental configuration. Each skier traverses the network through a sequence of downhill runs, lift queues, uphill transportation, and inter-lift

transfers with the duration of a traversal depending on the distance traveled and the mode of travel. The speed of each skier, with respect to mode, is held uniform for the simplicity of modeling.

Note: A reasonable addition to this simplification would be a dynamic movement system that scales skier speeds at the skill level or individually at the agent level.

2.5 Skier Decision-Making Under Incomplete Information

Upon entering the resort (spawning), each skier observes the waiting times of all operational lifts and stores these observations in memory.

Thereafter, the skier directly observes only the queue at their current location. The waiting times at other lifts constitute hidden states, represented by their most recently observed values.

Each individual skier *i* possesses its own kernel containing the last-observed queue size at every lift which serves terrain the skier may ski. Within this kernel, the skier *i* chooses to migrate and enter the line in which the sum of the travel time, from their current location, to that lift plus the last remembered queue time is the least.

$$j^{*}(t) = \underset{j \in A_i(t)}{arg\,min} (T_{travel\ c \to j} + k_j + T_{lift\ j\ ride\ time})$$

Observe the equation denotes the optimization function *arg min* as the summation of the travel time from lifts *c*, *c* being the arbitrary denotation of the lift the agent currently is at, to lift *j* plus the memory vector of lift line queue time at that lift *j* denoted by k_j. If multiple lifts share an equal estimated cost, the skier randomly chooses among tied admissible lifts.

Note a remembered observation is updated only when the skier revisits the corresponding lift. Consequently, the accuracy of stored information may deteriorate as the resort's actual queue

conditions change. This information staleness is a deliberate feature of the model. It permits decisions that are rational relative to the skier's available information to be suboptimal relative to the true state of the mountain.

Once a destination has been selected, the skier commits to that decision until reaching the destination. Decisions are reconsidered only at subsequent eligible decision points occurring after a descent of their chosen path has been completed.

2.6 Dynamic Lift-line Congestion

Queues are populated by the same skiers making the decisions described above changing the size and time required for a skier to board, ascend, and ultimately ski.

A skier's decision to remain at a lift contributes to its queue, while a decision to relocate changes the distribution of future arrivals across the resort. These changes influence subsequent waiting times and therefore the information upon which future decisions are based.

Now the constant-queue assumption is relaxed and treats each lift line as a dynamic population. Let $Q_j(t)$ denote the number of skiers waiting at lift j, $\lambda_j(t)$ its arrival rate, and $S_j(t)$ its boarding rate for that lift. Henceforth the population dynamics are governed by:

$$\frac{dQ_j}{dt} = \lambda_j(t) - S_j(t)$$

Where lift service is constrained by its maximum boarding capacity and the quantity of waiting skiers. Note that $0 \leq S_j(t) \leq \mu_j$, $S_j(t), \lambda_j(t) \geq 0$, and that the quantity of skiers cannot be negative.

Queue size, Q_j, dictates queue times, k, the time for an individual to arrive at the lift line before boarding, as a linear relationship with constant μ_j representing the fixed rate at which the

ski lift throughput at lift j. Note that $S_j(t)$ is not used, as a queue may be empty creating a divide by zero error.

$$k_j(t) = \frac{Q_j}{\mu_j}$$

Realized arrivals are independently decided by each agent, henceforth a uniform arrival formula is generally inappropriate. Note a simplified aggregate closure is required for analytical stability calculations.

Implementation

The model is implemented as a browser-based stochastic agent-based simulation, with a simulation engine separated from its visualization and interface. A fixed one-second time driver processes events at their scheduled times, including resort arrivals, movement completions, lift boardings, preference formation, and departure selections. A seeded pseudorandom number generator controls stochastic events and choices, allowing a run to be reproduced independently of animation speed. Each skier occupies one activity state at a time, with elapsed time accounted for as downhill skiing, queueing, uphill transportation, or transit.

The default operating day lasts 420 minutes and begins with an opening cohort of 650 skiers. Both values are configurable in the model. Two population modes are implemented. In fixed-population mode, the opening cohort circulates without additional arrivals or individual departures until closing. In arrival mode, the opening cohort is supplemented by a homogeneous Poisson arrival process during the first half of the day. At minute 210, new arrivals cease and an independent Poisson process selects existing skiers for departure at the same configured rate. The default rate is three events per minute. Departure selections are uniform among active skiers not

already selected, and an event has no effect when no eligible skier remains. Selection does not immediately remove a skier: a skier already queueing, riding, or descending completes the committed sequence before exiting at a safe base. A skier in base transit exits upon reaching the next base location. No additional random personal-stay departure mechanism is applied.

A configurable lodge-spawn probability determines each skier's entry location. With probability p_{lodge}, the skier enters at Main Lodge, positioned midway along the base promenade between L3 and L4. Otherwise, the entry location is sampled uniformly by physical distance over the union of base paths connecting that skier's admissible lifts. Shared path sections are counted once. Where only one lift is admissible, the spawn domain reduces to that lift's base. This rule applies to both opening skiers and subsequent arrivals. Travel from a sampled location or the lodge consumes simulated transit time and enters the initial lift-choice calculation.

Behavioral classes are assigned independently at entry using the proportions in Section 2.2; these are sampling probabilities rather than enforced population quotas. Within the lapper class, park, race, and general specialists are assigned with conditional weights of 10:2:5. Park lappers may choose between L1 and L2 using the common cost-minimization rule, then follow the designated park itinerary. Race lappers remain dedicated to L1 and the western race itinerary. General lappers receive one randomly assigned operational full run and its supporting lift, retaining that assignment throughout the day.

Mixed-skier preferences are implemented over lifts rather than fixed runs. At 40% of the time available between the skier's arrival and resort closing, the skier retains up to two experienced lifts with the lowest mean personally completed queue waits. A sole experienced lift remains the sole preference; unexperienced lifts are not added to fill the subset. If no queue wait

has been completed at the transition, preference formation is deferred until a subsequent eligible base decision with experience. Thereafter, the skier applies the common decision rule within the retained subset. Thus, these preferences restrict lift choice without prescribing two particular descents.

Each skier's queue memory is stored as lift-indexed queue counts and observation timestamps for all operating lifts, including lifts that may be inadmissible to that skier. Admissibility is enforced separately by filtering the candidate lift list before comparing costs. The opening cohort observes zero queues simultaneously before initial movements or queue admissions occur. Subsequent arrivals observe the then-current operating queues at entry. Thereafter, observations refresh only upon physical visits to corresponding lift bases. Passing an intermediate base updates its observation without permitting reconsideration of an already committed destination.

Movement follows explicit directed graph connections and the supplied route polylines. Geometric crossings do not create additional connections. The default speeds are eight metres per second on downhill runs, seven metres per second on ski traverses, and 2.2 metres per second on base transfers. Lift rides last seven minutes on L1 and L2 and 4.5 minutes on L3 and L4. Complete legal downhill itineraries are sampled with equal probability unless a lapper's specialization prescribes the itinerary. This differs from independently selecting an outgoing segment at each junction. Mandatory continuations preserve complete descents: the western park joins the lower portion of its associated black run, upper connectors continue through the appropriate lower terrain, and the race itinerary includes its return transfer.

Lift queues use first-in, first-out service. All four lifts currently have a configured maximum boarding capacity of 600 skiers per hour, implemented as one instantaneous individual boarding opportunity every six seconds. Empty boarding opportunities expire rather than accumulating. Actual throughput therefore depends on both capacity and available waiting skiers. At closing, no further arrivals or boardings occur. Queued skiers exit without boarding, while skiers already riding or descending finish their safe itineraries. These closing removals are recorded separately from boardings. Consequently, the exact discrete queue accounting is

$$Q□(t)=A□(t)-D□(t)-C□(t),$$

where A□(t) denotes cumulative queue admissions, D□(t) cumulative boardings, and C□(t) cumulative closing removals. Resort arrivals are distinct from queue admissions: a skier contributes to A□ only upon physically joining that queue.

Three operating configurations are provided: a three-lift baseline, a four-lift configuration retaining vacationers on L3, and a four-lift configuration permitting vacationers to use L3 or L4. Opening L4 also activates its associated terrain and dependent connectors. These comparisons therefore concern combined infrastructure, terrain, and access changes; they do not isolate a capacity effect alone. Operating configurations remain fixed within each run.

The interface displays live queue counts, estimated waits, completed waiting-time averages, boarding totals, skier states, and relocation counts. Queue-person-time and queue variability are calculated from queue changes rather than approximated solely from plotted samples. Waiting accumulated by skiers removed at closing remains included in total queue time but is excluded from completed-wait averages. Synchronized flat and interactive three-dimensional views display the same simulation state. The three-dimensional mountain is

schematic: its conical wrapping and interpolated route heights affect presentation only, without changing physical travel distances or decision costs.

Repeated-run functionality holds the selected parameters constant while varying the random seed. Individual queue-count or estimated-wait trajectories are overlaid using consistent colors for each lift. Plain-text and CSV exports provide settings, seeds, per-run summaries, and minute-by-minute observations. These outputs describe variation among simulation realizations; they do not perform forecasting or infer outcome probabilities.

Implementation checks cover reproducibility, population and time conservation, queue accounting, boarding capacity, legal movement, class-specific restrictions, spawning, departure rules, and preservation of incomplete queue information. The simulation supplies observations for examining congestion and aggregate queue behavior. However, the balance condition $\lambda \square = S \square$ is not treated as a stability result: a closed deterministic arrival representation and formal mathematical equilibrium-stability analysis remain separate analytical work. All numerical defaults and behavioral proportions are synthetic modeling assumptions rather than calibrated estimates for a particular resort.

References

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